

## Article

# How Far Can a U-Net Go? An Empirical Analysis of Music Source Separation Performance

Daniel Kostrzewa \* , Mikolaj Kondziolka, Robert Brzeski , Jeremiah Abimbola  and Pawel Benecki \* 

Department of Applied Informatics, Silesian University of Technology, 44-100 Gliwice, Poland

\* Correspondence: daniel.kostrzewa@polsl.pl (D.K.); pawel.benecki@polsl.pl (P.B.)

## Featured Application

This work supports the development of practical tools for isolating musical instruments and vocals from mixed audio. The optimized U-Net model can be integrated into applications for vocal extraction, remixing, audio restoration, and music education, as well as into music information retrieval systems requiring clean source signals.

## Abstract

Music source separation (MSS) focuses on decomposing a mixed audio signal into individual instrumental components and is increasingly relevant for music production, restoration, remixing, education, and music information retrieval. Deep learning methods, particularly U-Net architectures operating on time–frequency representations, have recently advanced the state of the art beyond traditional signal-processing techniques. This work presents an optimized multi-source U-Net model for separating selected musical instruments from stereo mixtures. The system uses magnitude spectrograms generated by the short-time Fourier transform and is trained and evaluated on the MUSDB18 dataset. We systematically examine architectural and training-related factors, including normalization strategies, dropout placement, optimizer selection, loss weighting, data augmentation, and spectrogram-domain modifications. Separation quality is measured using BSS Eval metrics, assessing artifacts, interference, and distortion. Experimental results show that the proposed configuration achieves competitive performance relative to established convolutional and U-Net-based open-source systems, especially in terms of vocal track separation, offering practical insights into designing efficient models for multi-instrument separation.

**Keywords:** music source separation; deep learning; U-Net; spectrogram processing; short-time Fourier transform (STFT); audio signal processing; convolutional neural networks; multi-task learning; BSS Eval; MUSDB18



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## 1. Introduction

Music source separation (MSS) is a task focused on decomposing a mixed audio signal into its individual components (instrumental or vocal). It is highly relevant in fields such as music production, audio restoration, remixing, music information retrieval, and musical education—everywhere where isolated source tracks are required. Beyond these domains, MSS is increasingly used in commercial audio applications, including automatic mixing assistance, karaoke generation, stem-based music recommendation, and adaptive music systems in gaming and virtual reality. The growing availability of multitrack recordings and large audio datasets further amplifies the importance of robust separation methods capable of operating in diverse acoustic conditions.

Early MSS algorithms relied on statistical models and signal-processing heuristics [1–3]: their performance was fundamentally limited by simplifying assumptions regarding the independence, sparsity, or stationarity of musical sources. Techniques such as Independent Component Analysis (ICA) [2], Non-negative Matrix Factorization (NMF) [3,4], and the Repeating Pattern Extraction Technique (REPET) [5] provided important conceptual foundations but struggled with polyphonic music, overlapping harmonics, and variations in timbre and production style. Moreover, these traditional methods typically required carefully designed preprocessing pipelines and were sensitive to parameter tuning, making them difficult to generalize across musical genres or recording environments.

Recent advances in deep learning have significantly improved the state of the art, enabling data-driven models to learn complex acoustic structures directly from large annotated datasets. Deep neural networks can model spectral patterns, temporal dependencies, and nonlinear interactions between sources, making them well-suited for the dense and highly correlated content present in musical mixtures. An important development in this area has been the shift from waveform-based to spectrogram-based modeling, where the separation task is framed as predicting a mask applied to the magnitude spectrogram of the mixture. Such mask-estimation approaches offer stability during training and exploit the structured nature of time–frequency representations. In deep learning, convolutional encoder–decoder architectures (like the U-Net) that work on time–frequency representations have proven to effectively separate multiple instruments [6,7]. These models use skip connections to maintain detailed spectral information and have yielded solid results on benchmarks like MUSDB18 [8]. Their hierarchical structure allows learning both local spectral textures—important for distinguishing harmonic instruments—and global context, which helps identify long-term temporal patterns characteristic of drums or vocals. However, their performance can vary significantly based on design decisions, data preprocessing, training methods, and ways to optimize for multiple sources [6]. Factors such as STFT window size, hop length, normalization strategies, loss functions, and balancing between instruments can substantially influence the resulting separation quality. Furthermore, multi-instrument setups pose challenges related to gradient competition between targets and the tendency of models to prioritize dominant sources in the mixture.

Because of these sensitivities, it is crucial to explore model and training components systematically to develop reliable and effective music source separation systems. Recent research also emphasizes the importance of data augmentation, including pitch shifting, time stretching, and mixing-based augmentation, to increase robustness and reduce overfitting. Additionally, post-processing techniques—such as Wiener filtering or multi-stage masking—can meaningfully improve final separation results by enforcing consistency between separated estimates and the input mixture. These aspects, often treated as implementation details, have a measurable cumulative effect on performance and generalization.

This study presents and optimizes a deep learning model capable of separating selected musical instruments from an audio mixture. The model is based on a multi-output U-Net architecture [9,10] trained on the MUSDB18 [8] dataset using magnitude spectrograms generated with the short-time Fourier transform (STFT). Building upon standard U-Net designs, the model incorporates several architectural and training variants inspired by recent literature and preliminary experiments conducted in this work. We focus on the analysis of the influence of architectural adjustments, training configurations, data augmentation, loss weighting strategies, and post-processing methods on the separation performance. In addition, attention is given to preprocessing choices such as STFT parameters and normalization schemes, which play an essential role in effective spectrogram modeling. Standard BSS Eval metrics [11] are used in comparisons, enabling quantitative assessment of artifacts, interference, and source distortion across different model vari-

ants. The resulting experimental study provides insights into which design elements most strongly affect performance in practical MSS scenarios.

### 1.1. Contribution

The main contribution of this work consists of a broad ablation study focusing on analyzing the limits of U-Net performance in an MSS task under realistic computational constraints, which are often treated as implementation details:

1. **Training modifications:** optimizers, normalization, regularization, and data augmentation;
2. **Architectural adjustments:** dropout configurations and STFT window variations;
3. **Post-processing techniques** including Wiener filtering.

That allows us to identify a high-performing configuration which is then compared with established open-source MSS systems.

### 1.2. Paper Structure

The remainder of the paper is organized as follows. Section 2 reviews the state of the art in music source separation. Section 3 describes methods and the dataset used in research. Section 4 presents the experimental results and analysis. Section 5 concludes the paper, describing limitations of the research and suggesting potential future research.

## 2. Related Work

Significant advances in music source separation (MSS) have been driven by increasingly sophisticated neural network architectures designed to capture both spectral and temporal characteristics of musical signals [12,13]. Early deep learning approaches in this period prominently featured encoder–decoder convolutional neural networks such as U-Net variants [14], including Conditioned U-Net (C-U-Net), which introduced control mechanisms via Feature-wise Linear Modulation (FiLM) layers to train a single generic model for multiple instrument separation tasks, achieving competitive performance relative to dedicated single-task models at reduced computational cost during training [15]. Wave-U-Net [16], an influential precursor operating in the time domain rather than the spectrogram domain, demonstrated end-to-end learning of audio source separation while preserving phase information, yielding separation quality comparable to state-of-the-art spectrogram-based approaches and establishing foundational principles for multi-scale feature extraction in source separation architectures. Multi-channel U-Net (M-U-Net) [9] further advanced this paradigm by employing weighted multi-task loss functions with Dynamic Weighted Average and Energy-Based Weighting strategies to address training bias arising from differing source energy levels, achieving SDR values of 4.89 dB for drums and 5.44 dB for bass on MUSDB18 without additional control parameters. Spleeter [17], released by Deezer in 2019 as a production-ready system, employed an encoder–decoder architecture trained on proprietary datasets to achieve state-of-the-art performance at that time, offering models for two-, four-, and five-stem separation with computational efficiency enabling real-time processing speed on GPUs, though the specific SDR metrics on MUSDB18 [8] were 5.73 dB for vocals and 4.02 dB for drums. These early convolutional approaches established baseline performance metrics while demonstrating the utility of multi-channel processing and architectural innovations over naïve masking-based methods.

The emergence of hybrid temporal–spectral domain processing and advanced convolutional architectures represented a crucial inflection point in MSS performance [18]. Open-Unmix (UMX) [19], a bidirectional LSTM-based architecture leveraging STFT representations with multichannel Wiener filtering post-processing, established baseline performance metrics on MUSDB18 with SDR values of 5.73 dB for vocals and 4.02 dB for accompaniment, providing a widely referenced reference implementation that enabled

systematic benchmarking. MMDenseLSTM [20], combining DenseNet-based convolutional feature extraction with multi-scale LSTM modules to capture both short- and long-range temporal dependencies, outperformed MMDenseNet [21] and single-scale LSTM architectures on DSD100 [22] and MUSDB18, achieving 6.62 dB SDR for drums and 4.93 dB for drums/bass separation with substantially reduced parameters relative to simple architectural combinations. D3Net [23] introduced densely connected multi-dilated convolutions that simultaneously modeled multiple temporal resolutions within single convolutional layers, achieving state-of-the-art performance of 6.01 dB average SDR through rapid receptive field growth while avoiding aliasing artifacts inherent to naive dilated convolution integration in dense architectures. Demucs [24], developed by Facebook AI Research, pioneered hybrid waveform/spectrogram domain source separation by operating parallel temporal and spectral branches within a unified U-Net encoder-decoder framework, achieving 6.84 dB SDR for vocals when trained on MUSDB18-HQ [25] with augmented training data, with subsequent Hybrid Transformer variants incorporating cross-domain attention mechanisms to achieve 9.20 dB SDR when augmented with 800 additional training songs and per-source fine-tuning with sparse attention. These hybrid and parameter-efficient approaches demonstrated substantial improvements over single-domain processing, suggesting that multi-representation learning provides complementary modeling capabilities for capturing frequency-dependent and phase-dependent source characteristics.

The band-splitting paradigm became an important architectural innovation between 2022 and 2024. Band-Split RNN (BSRNN) [26] showed that separating frequency bands based on the bandwidth needs of each source can significantly improve separation performance. BSRNN outperformed the Music Demixing Challenge 2021 benchmarks, achieving SDR values above 8.0 dB on vocal tracks. It reached these results by using dedicated dual-path RNN modules. One module handled sequence-level features, and the other processed band-level features. BSRNN then applied time–frequency mask estimation to produce the final separation. This approach was also more computationally efficient than processing the full frequency domain at once. The findings suggest that allocating resources to specific frequency bands leads to better separation quality. Band-Split RoPE Transformer (BS-RoFormer) extended this idea by combining band-split processing with a hierarchical Transformer architecture that used Rotary Position Embedding [27]. It won the Sound Demixing Challenge 2023 and achieved an average SDR of 11.99 dB when trained on MUSDB18-HQ plus 500 additional songs. The Mel-Band RoFormer [28] further refined the band-split approach. It used mel-scale frequency projections to create overlapping sub-bands that better match human auditory perception. This model outperformed non-overlapping band-split systems across vocal, drum, and accompaniment separation tasks.

Lightweight variants, including parameter-efficient band-split attention models, achieved approximately 10 dB SDR with substantially reduced parameter counts compared to full transformer variants, democratizing access to high-performance separation architectures [29]. Recent work on dual-path temporal-frequency convolution and time-distributed fully connected UNet (DTTNet) [30,31] achieved 10.12 dB cSDR for vocal separation while employing 86.7% fewer parameters than BSRNN, demonstrating that architectural design choices may yield equivalent or superior performance compared to scale-based improvements alone. Critically, recent methodological analysis demonstrated that reported SDR improvements of 0.22 to 0.37 dB can be attributed solely to increased inference-time overlap rather than architectural differences [32], indicating that performance comparisons require careful methodological consistency and that boundary artifact reduction may constitute a substantial portion of reported improvements across contemporary architectures [33]. The primary limitations of contemporary architectures include reduced modeling capac-

ity when separation challenges exceed architectural assumptions regarding frequency-domain properties, reduced effectiveness on limited training data, and computational overhead during training requiring substantial GPUs and extended training periods for optimal convergence.

### 3. Materials and Methods

#### 3.1. Dataset

The MUSDB18 dataset [8] was used for training and evaluation. MUSDB18 is one of the most widely adopted benchmarks for music source separation research. It contains a total of 150 professionally produced music tracks, corresponding to approximately 10 h of audio material. The dataset is explicitly divided by its authors into two subsets to support supervised machine learning workflows: the training set includes 100 tracks with a total duration of 6.5 h, while the test set consists of 50 tracks with a combined duration of 3.5 h. Such a predefined split provides a possibility to directly compare different methods of track separation.

All tracks in MUSDB18 are provided in a multi-track format, which allows several time-aligned audio streams to be stored in a single file. Each file contains the complete mixture as well as four isolated reference sources: *drums*, *bass*, *other* (accompaniment), and *vocals*. The mixture is an exact linear sum of the four stems, ensuring that the dataset provides reliable and fully consistent ground truth for supervised source separation tasks. All signals are stored in stereo at a sampling rate of 44.1 kHz.

MUSDB18 was prepared specifically for the 2018 SiSEC campaign [34] focused on audio source separation. It is an aggregated dataset composed of several underlying collections: 100 tracks originate from the DSD100 dataset [22], 46 from MedleyDB [35], 2 were contributed by Native Instruments, and 2 by the Canadian band The Easton Ellises. The use of MUSDB18 ensures that the results obtained in this work remain directly comparable to prior studies and published baselines.

This study uses the standard compressed version of MUSDB18 [8], rather than the MUSDB18-HQ variant [25], in order to maintain compatibility with the majority of existing literature and reported benchmarks.

#### 3.2. Preprocessing

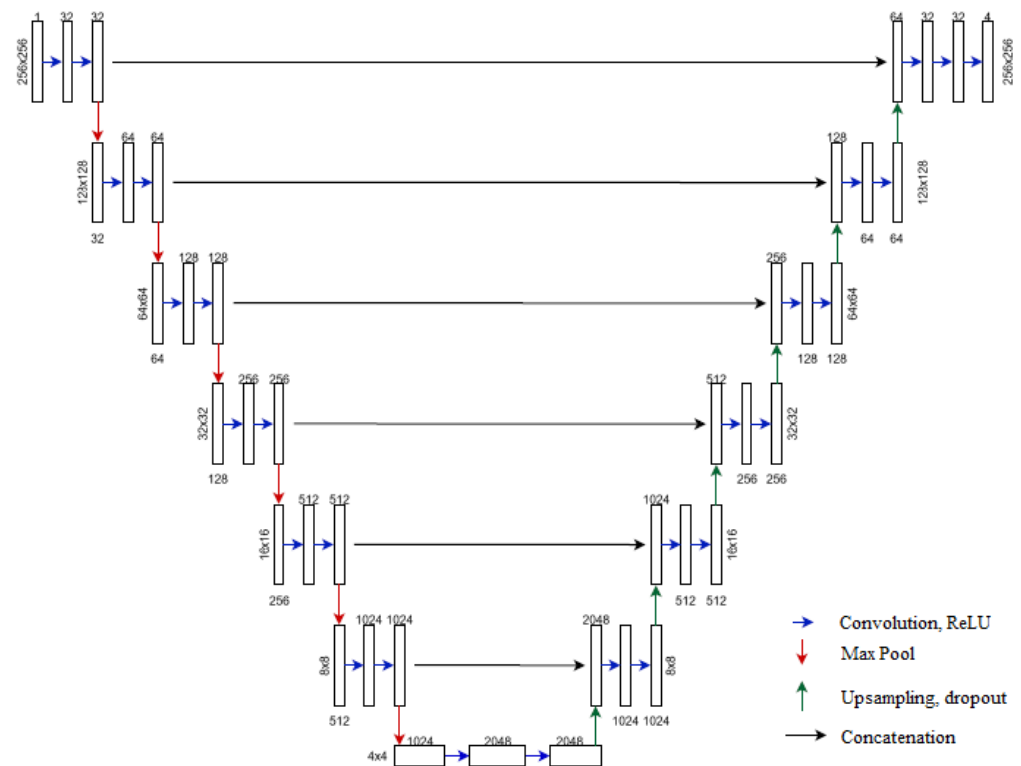
After loading the song, channels are fused through averaging to obtain a single signal. Then, the signal frequency was reduced to 10,880 Hz. The song was then divided into non-overlapping fragments of 6 s each, and each of these fragments was subjected to STFT using a Hanning window [36] of size 1022 and a hop size of 256 samples. This step transforms the signal from the time domain to a time–frequency representation. The spectrogram of each fragment has dimensions  $512 \times 256$ . To cover the entire song, the last fragment was cut from the end of the song. The magnitude was extracted from STFT and serves as the input to the network. This procedure was applied to each song. For each processed fragment, the following is saved: the magnitudes of the full song, magnitudes of individual channels, and the phase of the full song—which is used in the reconstruction step.

#### 3.3. Model Architecture

The proposed system is based on a U-Net architecture, which is frequently used for the track separation task [6,37]. The structure is presented in Figure 1. The encoder part consists of six convolutional blocks (made of two convolution layers, each followed by a ReLU activation function). The number of channels increases progressively (from 32 to 1024), while reducing dimensionality ( $256 \times 256$  down to  $8 \times 8$ ) at each next depth level of the encoder. The bottleneck consists of two convolutions of dimensionality  $4 \times 4$  and



2048 channels. The decoder mirrors the encoder, using transposed convolutions for up-sampling. Connections between corresponding layers of the encoder and decoder are present. The output layer in the encoder predicts four time–frequency masks, one for each target source.



**Figure 1.** Diagram of the designed network.

### 3.4. Mask Estimation and Loss Function

The proposed M-U-Net model is trained to simultaneously generate real-valued masks for each of the target sources in a multi-task manner. These masks, denoted as  $\tilde{M}_i(n, m)$  for source  $i$ , are element-wise multiplied with the magnitude spectrogram of the input mixture  $S_{\text{mix}}(n, m)$  to produce the estimated spectrogram  $\hat{S}_i(n, m) = \tilde{M}_i(n, m) \cdot S_{\text{mix}}(n, m)$ .

Rather than computing the loss directly on the mask errors, an indirect loss approach is adopted, which minimizes the discrepancy at the level of the reconstructed spectrograms. For each source  $i$ , the loss  $\mathcal{L}_i$  is defined as the L1 norm between the estimated and ground-truth magnitude spectrograms:

$$\mathcal{L}_i = \sum_{n=1}^T \sum_{m=1}^F |S_i(n, m) - \tilde{M}_i(n, m) \cdot S_{\text{mix}}(n, m)|, \quad (1)$$

where  $T$  is the number of time frames,  $F$  is the number of frequency bins, and  $S_i(n, m)$  is the ground-truth magnitude spectrogram for source  $i$ .

This indirect loss is motivated by findings in related fields such as speech denoising [38], as it aligns more closely with the ultimate goal of producing separated signals that match the target sources. Additionally, it implicitly weights mask errors by the mixture's amplitude, emphasizing precision in high-energy regions where artifacts are more perceptible, unlike direct mask-based losses that treat all errors equally regardless of amplitude.

To mitigate the imbalance in energy distributions across sources, which could otherwise bias the model toward high-energy instruments like drums at the expense of quieter

ones like bass, an Energy-Based Weighting (EBW) strategy is employed. The weight for each source  $i$  is computed dynamically within each training batch as:

$$w_i(t) = \frac{\max_{j \in \{1, \dots, K\}} E_j(t)}{E_i(t)}, \quad (2)$$

where  $E_i(t)$  is the average energy of source  $i$  in the current batch  $t$ , defined as the mean of the squared magnitudes in its spectrogram, and  $K$  is the number of sources.

The total loss is then the weighted sum:

$$\mathcal{L} = \sum_{i=1}^K w_i \cdot \mathcal{L}_i. \quad (3)$$

This weighting ensures that the model allocates equal attention to all sources by normalizing the contribution of each loss term relative to the highest-energy source in the batch, preventing dominance by louder instruments and promoting balanced learning.

### 3.5. Training Procedure

The training length limit was set to 2000 epochs or 24 h of duration (whichever occurs first). Batch size was set to 64 samples, where each sample contains the amplitude of the full song and separate amplitudes of the sources. Before feeding the data to the input, they were sampled to a logarithmic scale, resulting in spectrograms of dimensions  $256 \times 256$ . The weighted loss function described in Section 3.4 and the Adam optimizer with a learning rate of 0.001 were used for training (after experimentation, replacing the initial mini-batch SGD with LR 0.01). The loss calculation process was performed on spectrograms in linear scale (after initial experiments with log scale). The MUSDB18 dataset was split into training (100 songs) and test (50 songs) sets as defined in the dataset itself. Evaluation was performed on the test set to assess model performance. Additional hyperparameters include L2 weight regularization with a coefficient of 0.01 and dropout rate of 0.5 in the first layers of the decoder. Data augmentation techniques applied include pitch shifting and time–frequency masking with spectrogram averages.

### 3.6. Audio Reconstruction

The song reconstruction process was performed on spectrograms corresponding to six-second parts of the audio track. Fragments of a given song are sampled to a logarithmic scale and fed sequentially to the input. The masks generated by the model were then sampled to a linear scale to obtain spectrograms of dimensions  $512 \times 256$ . After processing, the masks were multiplied by the amplitude and phase of the song, then the obtained spectrograms were subjected to Inverse STFT, resulting in the audio tracks of the separated instruments. Since the model operates on six-second fragments, the tracks obtained during reconstruction were combined to recreate the full instrument track from the given song. The Wiener filter (single iteration) was applied in the final model as a post-processing step to further improve separation quality by recalculating masks based on power ratios, enhancing SIR at the cost of slight SAR reduction.

### 3.7. Evaluation Metrics

The quality of the separated audio signals is assessed using the BSS Eval metrics, a standard framework for evaluating blind source separation algorithms introduced in [11]. These metrics provide an objective, quantitative measure of separation performance by decomposing the estimated signal and computing ratios that capture different aspects of separation errors. The three primary metrics employed are the **Signal-to-Distortion**

**Ratio (SDR), Signal-to-Interference Ratio (SIR), and Signal-to-Artifacts Ratio (SAR)**, all expressed in decibels (dB).

**SDR** is the most comprehensive metric, evaluating the overall separation quality by measuring the ratio of the energy of the correctly extracted target signal to the combined energy of all errors, including interferences and artifacts. A **high SDR value** indicates that the separated signal is both **free from leakage** of other sources and **minimally distorted** by the algorithm. It is defined as:

$$\text{SDR} = 10 \log_{10} \frac{\|S_{\text{target}}\|^2}{\|e_{\text{interf}} + e_{\text{artif}} + e_{\text{noise}}\|^2}, \quad (4)$$

where  $S_{\text{target}}$  is the target component of the estimated signal, and  $e_{\text{interf}}$ ,  $e_{\text{artif}}$ , and  $e_{\text{noise}}$  represent errors due to interference, artifacts, and noise, respectively.

**SIR** focuses on the model's ability to suppress unwanted sources, quantifying the ratio of the target signal energy to the interference from other instruments. It ignores algorithmic artifacts, emphasizing isolation from cross-talk. A **high SIR** signifies **minimal flow from other tracks** into the separated source. The formula is:

$$\text{SIR} = 10 \log_{10} \frac{\|S_{\text{target}}\|^2}{\|e_{\text{interf}}\|^2}. \quad (5)$$

**SAR** evaluates the distortions introduced solely by the separation algorithm, independent of interference from other sources. It measures the ratio of the energy of the target signal plus allowable errors (interference and noise) to the energy of artifacts. A **high SAR** indicates **natural-sounding** output with **low musical noise**. It is computed as:

$$\text{SAR} = 10 \log_{10} \frac{\|S_{\text{target}} + e_{\text{interf}} + e_{\text{noise}}\|^2}{\|e_{\text{artif}}\|^2}. \quad (6)$$

These metrics are computed using the `museval` library, which implements BSS Eval version 4, ensuring consistent and reproducible evaluation across the MUSDB18 test set.

### 3.8. Implementation Details

The experiments were conducted on the Google Colab environment, using an NVIDIA L4 GPU with 24 GB VRAM and 53 GB system RAM. The dataset was stored in RAM to accelerate data loading. The implementation used Python 3.11.13. Neural networks were built and trained with PyTorch 2.6.0, supporting dynamic graphs and GPU acceleration via CUDA. Dataset preprocessing and audio reconstruction employed Librosa 0.11 for STFT and Inverse STFT. Stem format files were handled using stempeg 0.2.4, interfacing with FFmpeg. Separation quality was evaluated with `mir_eval` 0.8.2, providing BSS Eval metrics. NumPy 2.0.2 served as the foundation for array operations across libraries.

## 4. Experimental Evaluation

This section describes the experiments and their results. The results of each experiment are the median (Mdn), mean (M), and standard deviation (SD) of the SDR, SAR, and SIR metrics. For the calculation of the metrics, reference tracks were downsampled to 10,880 Hz and converted to mono to match the dimensions of the tracks generated by the model.

### 4.1. Results Obtained Using the Baseline Model

Table 1 shows the results obtained by the baseline model designed and trained as described in Section 3.3.

The model excels at separating vocals, which achieve the highest SDR (Median 4.85 dB) and SIR (Median 10.77 dB) metrics, meaning that the remaining instruments



penetrate the isolated track only slightly. Drum separation also yielded good results, but the model performed poorly in separating the bass from the other accompaniment. The bass achieved very low SDR (Median 1.79 dB) and SIR (Median 3.89 dB) metrics. The other accompaniment is the most difficult track to separate, as evidenced by the lowest SDR (Median 1.57 dB) and average SIR (Median 5.33) and SAR (Median 5.41) scores.

In most cases, the mean is lower than the median. This suggests that there are songs in the test set on which the model performs exceptionally poorly, and these extremely negative results lower the mean. The median is a more reliable indicator of typical model performance. The standard deviation values are very high, often close to or even greater than the mean, meaning that model performance depends on the specific song.

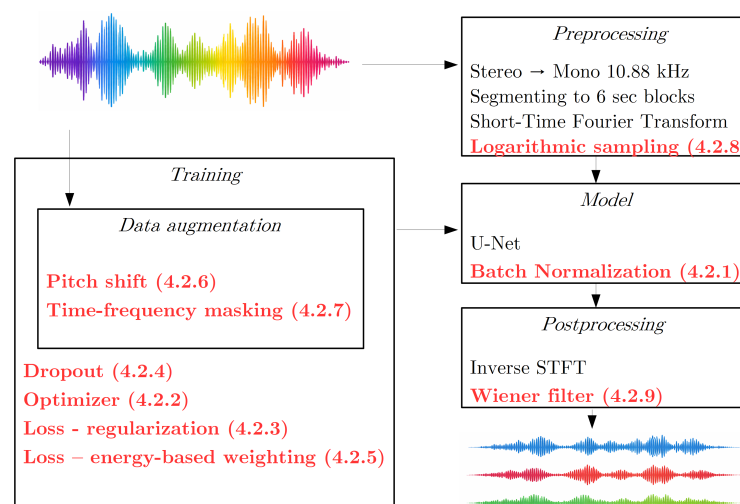
**Table 1.** Obtained results of the baseline model.

| Instrument     | SDR [dB] |      |      | SAR [dB] |      |      | SIR [dB] |       |      |
|----------------|----------|------|------|----------|------|------|----------|-------|------|
|                | Mdn      | M    | SD   | Mdn      | M    | SD   | Mdn      | M     | SD   |
| Drums          | 3.63     | 3.86 | 3.47 | 5.99     | 6.81 | 3.14 | 8.14     | 8.37  | 3.78 |
| Bass           | 1.79     | 1.46 | 4.23 | 7.02     | 6.87 | 3.24 | 3.89     | 4.03  | 4.51 |
| Others         | 1.57     | 0.74 | 3.33 | 5.41     | 4.53 | 3.35 | 5.33     | 4.94  | 2.70 |
| Vocal          | 4.85     | 4.10 | 4.70 | 6.53     | 6.11 | 3.89 | 10.77    | 10.51 | 4.49 |
| Entire dataset | 2.99     | 2.54 | 4.23 | 6.20     | 6.08 | 3.55 | 7.01     | 7.03  | 4.68 |

#### 4.2. Architectural and Training Adaptations

To improve the obtained results, various modifications to the baseline model were iteratively introduced in subsequent steps. Table 2 lists the modifications used to build the final model and their impact on the overall separation quality. Due to the discrete nature of the tested hyperparameter values, the results are presented in tabular form rather than as continuous parameter sweeps.

Figure 2 illustrates the system architecture, highlighting in red the components and stages where architectural and training modifications were applied. This overview helps contextualize the subsequent experimental adjustments and their impact on performance.



**Figure 2.** Overview of the system, highlighting in red the points where architectural and training modifications were applied. Sections describing the modifications and corresponding results are referenced in brackets.

**Table 2.** Collected modifications to the baseline model.

| Modifications  | SDR [dB] |      |      | SAR [dB] |      |      | SIR [dB] |       |      |
|----------------|----------|------|------|----------|------|------|----------|-------|------|
|                | Mdn      | M    | SD   | Mdn      | M    | SD   | Mdn      | M     | SD   |
| Baseline model | 2.99     | 2.54 | 4.23 | 6.20     | 6.08 | 3.55 | 7.01     | 7.03  | 4.68 |
| Normalization  | 3.22     | 2.97 | 4.90 | 6.96     | 6.82 | 3.50 | 6.88     | 6.99  | 4.92 |
| Adam Optimizer | 3.45     | 3.15 | 4.31 | 6.91     | 6.67 | 3.43 | 7.34     | 7.59  | 5.04 |
| Regularization | 3.54     | 3.22 | 4.18 | 6.87     | 6.67 | 3.42 | 7.46     | 7.67  | 4.86 |
| Dropout        | 3.50     | 3.25 | 4.14 | 6.82     | 6.67 | 3.32 | 7.63     | 7.72  | 4.91 |
| Loss Weighting | 3.70     | 3.36 | 4.20 | 6.90     | 6.75 | 3.35 | 7.75     | 7.91  | 4.98 |
| Augmentation 1 | 3.73     | 3.49 | 4.11 | 7.20     | 6.79 | 3.46 | 7.80     | 8.13  | 4.09 |
| Augmentation 2 | 3.91     | 3.80 | 3.93 | 7.21     | 7.21 | 3.15 | 7.86     | 8.20  | 4.92 |
| Sampling       | 4.40     | 4.17 | 4.02 | 7.29     | 7.90 | 3.28 | 9.41     | 9.27  | 5.20 |
| Wiener Filter  | 4.54     | 4.42 | 4.13 | 6.46     | 6.32 | 3.49 | 12.20    | 11.59 | 5.59 |

#### 4.2.1. Batch Normalization

The first modification to the baseline model was the introduction of batch normalization.

The obtained results clearly indicate that the use of batch normalization significantly improves separation quality compared to the baseline model without normalization. As can be seen in the summary (Table 2), the introduction of this technique increased the overall median SDR metric from 2.99 dB to 3.22 dB and the SAR from 6.20 dB to 6.96 dB. This means that the model generates separations of higher overall quality. Interestingly, the median SIR decreased slightly (from 7.01 dB to 6.88 dB), suggesting that the main benefit comes from distortion reduction and not necessarily from better suppression of other sources.

#### 4.2.2. Optimizer

The research carried out at this stage examined the influence of four optimizers on the obtained separation results. The results obtained are presented in Table 3. Although the differences are small, there are clear favorites. Adam provides the most balanced results for the overall separation task. The results in Table 3 indicate that Adam achieved the highest median SDR (3.45 dB) and SIR (7.34 dB). SCION also delivered good results, while the other optimizers tested delivered poorer outcomes. At the same time, the analysis of Table 2 shows that the introduction of this modification in most cases improved the quality of separation compared to the previous models. Therefore, further studies used the Adam optimizer.

**Table 3.** The influence of optimizers on the overall separation quality.

| Optimizer | SDR [dB] |      |      | SAR [dB] |      |      | SIR [dB] |      |      |
|-----------|----------|------|------|----------|------|------|----------|------|------|
|           | Mdn      | M    | SD   | Mdn      | M    | SD   | Mdn      | M    | SD   |
| SGD       | 3.22     | 2.97 | 4.90 | 6.96     | 6.82 | 3.50 | 6.88     | 6.99 | 4.92 |
| ADAM      | 3.45     | 3.15 | 4.31 | 6.91     | 6.67 | 3.43 | 7.34     | 7.59 | 5.04 |
| LAMB      | 3.22     | 2.96 | 4.13 | 6.69     | 6.54 | 3.27 | 7.12     | 7.37 | 4.98 |
| SCION     | 3.42     | 3.12 | 4.12 | 6.84     | 6.70 | 3.33 | 7.29     | 7.49 | 4.89 |

#### 4.2.3. Regularization

Weight regularization is a technique used to prevent model overfitting. It works by adding an additional term to the loss function that is proportional, according to a regularization coefficient, to the sum of the squares of all the weights in the neural network.

The results obtained after introducing this modification are presented in Table 4. They indicate that for overall separation quality, regularization with a factor of 0.01 proved to be the most beneficial. For this value, SDR achieved the highest median SDR (3.54 dB) and SIR (7.46 dB). This means that the model with this regularization produces separations of better overall quality and more effectively suppresses interference from other instruments compared to the baseline model. A regularization factor of 0.001 also resulted in improvements, but at a slightly lower level. L2 weight regularization is a beneficial technique for this separation model. Its main advantage is its improved ability to isolate individual sources from each other, but this comes at the cost of introducing minor artifacts. The overall balance, measured by the SDR metric, is positive. Therefore, further studies used a regularization factor of 0.01.

**Table 4.** The influence of L2 weight regularization on the overall separation quality.

| Coefficient | SDR [dB] |      |      | SAR [dB] |      |      | SIR [dB] |      |      |
|-------------|----------|------|------|----------|------|------|----------|------|------|
|             | Mdn      | M    | SD   | Mdn      | M    | SD   | Mdn      | M    | SD   |
| 0—None      | 3.45     | 3.15 | 4.31 | 6.91     | 6.67 | 3.43 | 7.34     | 7.59 | 5.04 |
| 0.01        | 3.54     | 3.22 | 4.18 | 6.87     | 6.67 | 3.42 | 7.46     | 7.67 | 4.86 |
| 0.001       | 3.53     | 3.20 | 4.31 | 6.94     | 6.74 | 3.30 | 7.35     | 7.67 | 5.24 |

#### 4.2.4. Dropout

The baseline model uses a dropout factor of 0.1 for each decoder layer. Based on the U-Net architecture described in Cohen-Hadria et al. [39], a dropout factor of 0.5 was used for the first three decoder layers.

The results obtained are presented in Table 5. Analysis of the results shows that this modification resulted in an overall improvement in separation quality. Increasing the dropout factor to 0.5 increased the median SDR from 3.45 dB to 3.50 dB and the median SIR from 7.34 dB to 7.63 dB. This indicates that increased dropout leads to better overall separation quality and more effective interpath interference suppression. Therefore, further studies used Dropout factor 0.5 in the first 3 layers of the decoder.

**Table 5.** The influence of increasing the dropout factor in the initial decoder layers on the overall separation quality.

| Dropout Factor | SDR [dB] |      |      | SAR [dB] |      |      | SIR [dB] |      |      |
|----------------|----------|------|------|----------|------|------|----------|------|------|
|                | Mdn      | M    | SD   | Mdn      | M    | SD   | Mdn      | M    | SD   |
| 0.1            | 3.45     | 3.15 | 4.31 | 6.91     | 6.67 | 3.43 | 7.34     | 7.59 | 5.04 |
| 0.5            | 3.50     | 3.25 | 4.14 | 6.82     | 6.67 | 3.32 | 7.63     | 7.72 | 4.91 |

#### 4.2.5. Loss Weighting

The next modification is based on the work of Venkatesh Kadandale et al. [9]. EBW P1 assumes constant weights calculated based on the average energy of each source across the entire dataset. The weight for a given source is inversely proportional to its energy, which means that lower-energy sources receive a higher weight. EBW P2 works similarly, but uses the square of the energy ratio to calculate the weights, which is intended to further strengthen the weights for lower-energy sources. In EBW InstP1, the

weights are dynamically calculated for the current dataset; this strategy was initially implemented in the model. Based on this strategy, EBW InstP2 was introduced, which dynamically calculates based on the squared energies. This strategy was not presented in the original article.

The results obtained are presented in Table 6. After analyzing this table, we can see that for overall separation quality, the EBW InstP2 strategy improved results in all metrics, achieving a median SDR of 3.70 dB, SAR of 6.90 dB and SIR of 7.75 dB. Therefore, further studies used the EBW InstP2 loss weighting strategy.

**Table 6.** The influence of changing the loss weighting strategy on the overall separation quality.

| Strategy   | SDR [dB] |      |      | SAR [dB] |      |      | SIR [dB] |      |      |
|------------|----------|------|------|----------|------|------|----------|------|------|
|            | Mdn      | M    | SD   | Mdn      | M    | SD   | Mdn      | M    | SD   |
| EBW InstP1 | 3.50     | 3.25 | 4.14 | 6.82     | 6.67 | 3.32 | 7.63     | 7.72 | 4.91 |
| EBW P1     | 3.44     | 3.20 | 4.19 | 6.84     | 6.66 | 3.36 | 7.40     | 7.67 | 4.99 |
| EBW P2     | 3.55     | 3.23 | 4.14 | 6.88     | 6.66 | 3.39 | 7.53     | 7.82 | 5.07 |
| EBW InstP2 | 3.70     | 3.36 | 4.20 | 6.90     | 6.75 | 3.35 | 7.75     | 7.91 | 4.98 |

#### 4.2.6. Augmentation 1: Pitch Shift

Data augmentation is the process of artificially expanding a training set by modifying existing data. This study employed a pitch shift operation, which involves shifting spectrogram samples along the frequency axis. This is a simplified method, but it accomplishes the purpose of data augmentation. Songs raised by one semitone were added to the training set.

Pitch shifting as an augmentation method improved the overall model results, as shown in Table 2. The median overall SDR increased from 3.70 dB to 3.73 dB, and the SAR metric, which measures the amount of artifacts, improved from 6.90 dB to 7.20 dB. Therefore, further research used this modification in the learning process.

#### 4.2.7. Augmentation 2: Time–Frequency Masking

Another technique to extend the dataset is time–frequency masking, described in the work of Helin Wang et al. [40]. This technique involves masking rectangular regions of the spectrogram, which improves model generalization.

Three variants of this technique, presented in the paper, were tested in this study. Zero masking is the simplest method, in which a selected fragment of the spectrogram is replaced with zeros. Masking with another spectrogram involves replacing a selected fragment of the target spectrogram with its corresponding fragment from another spectrogram. The final variant is mean masking, where the selected fragment is replaced with the arithmetic mean of the original fragment and its corresponding fragment from another spectrogram. This variant introduces less noise than complete replacement with another spectrogram.

The results obtained are presented in Table 7. The best strategy was masking the average of the spectrograms. This method achieved the best results in SDR (median 3.91 dB) and SAR (median 7.21 dB). Therefore, further research used this modification in the learning process.

**Table 7.** The influence of of the time–frequency mask on the overall separation quality.

| Type of Mask        | SDR [dB] |      |      | SAR [dB] |      |      | SIR [dB] |      |      |
|---------------------|----------|------|------|----------|------|------|----------|------|------|
|                     | Mdn      | M    | SD   | Mdn      | M    | SD   | Mdn      | M    | SD   |
| No Mask             | 3.73     | 3.49 | 4.11 | 7.20     | 6.79 | 3.46 | 7.80     | 8.13 | 4.09 |
| Zero Masking        | 3.84     | 3.53 | 4.23 | 7.13     | 6.94 | 3.52 | 7.90     | 8.10 | 5.04 |
| Another spectrogram | 3.77     | 3.60 | 4.13 | 7.15     | 6.98 | 3.31 | 7.84     | 8.15 | 5.90 |
| Average of spect.   | 3.91     | 3.80 | 3.93 | 7.21     | 7.21 | 3.15 | 7.86     | 8.20 | 4.92 |

#### 4.2.8. Sampling—Modifying Sampling to a Logarithmic Scale

During model training, spectrograms with a linear frequency scale are subjected to nonlinear sampling, which transforms the frequency axis to a logarithmic scale. This study compares three spectrogram sampling strategies. The baseline approach involved sampling both the input and reference data to a logarithmic scale. The neural network processes the data and outputs masks in the same logarithmic space, where the loss is then calculated. Next, the performance of a model operating entirely on linear-scale spectrograms was tested and explored a hybrid approach in which the network operates on logarithmic-scale data, but its output is downsampled to the original linear frequency scale before calculating the loss. Therefore, the loss is calculated by comparing the reconstructed linear spectrograms.

Analysis of the results obtained in Table 8 shows that using logarithmic sampling is very reasonable. The results indicate that operating on a linear scale [No log. scale] sometimes produces negative metric values, meaning that the noise exceeds the expected signal. The best results were obtained with the second method [On the original shape], where the loss is calculated based on the original shape of the spectrograms. This approach, which calculates the loss on a linear frequency scale, yields better results than calculating it on a logarithmic scale; for overall separation quality, the median SDR increased from 3.82 dB to 4.40 dB. The neural network learns better in logarithmic space, but the ultimate goal is to reconstruct the signal on a linear scale. By calculating the loss on the target scale, a more accurate error value can be determined. Therefore, further studies used this modification.

**Table 8.** The influence of modifying sampling to a logarithmic scale on the overall separation quality.

| Modification Type    | SDR [dB] |       |      | SAR [dB] |      |      | SIR [dB] |       |      |
|----------------------|----------|-------|------|----------|------|------|----------|-------|------|
|                      | Mdn      | M     | SD   | Mdn      | M    | SD   | Mdn      | M     | SD   |
| Loss on a log. scale | 3.82     | 3.71  | 4.05 | 7.14     | 7.05 | 3.32 | 8.21     | 8.23  | 4.90 |
| On the orig. shape   | 4.40     | 4.17  | 4.02 | 7.29     | 7.90 | 3.28 | 9.41     | 9.27  | 5.20 |
| No log. scale        | −4.33    | −4.66 | 4.11 | 3.64     | 4.08 | 3.58 | −1.42    | −1.57 | 5.33 |

#### 4.2.9. Wiener Filter

The Wiener filter is a post-processing technique used to further improve separation quality. The filter operates by generating a new mask based on the ratio of the desired instrument power to the sum of all instrument powers at each point in the spectrogram [41]. The implementation available in the norbert library [<https://sigsep.github.io/norbert> (accessed on 12 December 2025)] was used for this study. The study also examined the effect of two consecutive filter iterations, where the output of the first is the input to the second.

The results obtained are presented in Table 9. As can be seen, applying the Wiener filter significantly improved path separation. However, the Wiener filter introduces new distortions into the signal, which manifests itself in a significant drop in the SAR met-

ric. Applying two filter iterations exacerbates this phenomenon, increasing the SAR, but creating so many artifacts that the overall quality measured by the SDR metric begins to decline. Ultimately, applying a single iteration of the Wiener filter yielded the greatest improvement; this variant achieved the best balance between suppressing interference sources and introducing artifacts. Therefore, this modification was used.

**Table 9.** The influence of the Wiener filter on the overall separation quality.

| Filter Type         | SDR [dB] |      |      | SAR [dB] |      |      | SIR [dB] |       |      |
|---------------------|----------|------|------|----------|------|------|----------|-------|------|
|                     | Mdn      | M    | SD   | Mdn      | M    | SD   | Mdn      | M     | SD   |
| No Filter           | 4.40     | 4.17 | 4.02 | 7.29     | 7.90 | 3.28 | 9.41     | 9.27  | 5.20 |
| Wiener Filter       | 4.54     | 4.42 | 4.13 | 6.46     | 6.32 | 3.49 | 12.20    | 11.59 | 5.59 |
| Wiener—2 Iterations | 4.20     | 4.13 | 4.14 | 5.89     | 5.75 | 3.56 | 12.72    | 12.21 | 5.73 |

#### 4.3. Final Model Results

The modifications described in Section 4.2 were used to build the final model. The starting point was the baseline model, to which successive modifications were iteratively introduced to improve the separation quality. The results obtained are presented in Table 10. Comparing these results with those obtained for the baseline model (Table 1), it can be seen that most values have improved significantly.

#### Characteristics of Challenging Test Cases

To better understand the limitations of the final model, we analyzed the tracks ranked low by SDR across all source categories. Several common characteristics emerged among poorly separated tracks. The model struggles significantly with heavily processed or synthetic sounds: tracks featuring vocalized effects, distorted instruments (particularly in metal and hard rock genres), and synthesizer-generated sources consistently yielded low SDR values. Genre-specific challenges were also apparent—reggae and hip-hop tracks with sparse arrangements proved difficult to separate, likely due to the model’s reliance on continuous spectral features. Electronic music with loop-based production and minimal traditional instrumentation presented similar difficulties. Analysis of the MUSDB18 training set revealed underrepresentation of these challenging cases: two tracks with vocalization, one reggae track (only 17 s long), two hip-hop tracks, three electronic tracks, and limited heavily distorted metal content. These findings suggest that the observed performance variance stems primarily from genre imbalance in the training data and the model’s reduced capacity to handle non-acoustic, heavily processed audio sources. This suggests directions for dataset enhancement in future work.

**Table 10.** Obtained results of the final model.

| Instrument     | SDR [dB] |      |      | SAR [dB] |      |      | SIR [dB] |       |      |
|----------------|----------|------|------|----------|------|------|----------|-------|------|
|                | Mdn      | M    | SD   | Mdn      | M    | SD   | Mdn      | M     | SD   |
| Drums          | 5.66     | 5.99 | 3.63 | 6.74     | 7.14 | 3.52 | 14.91    | 14.61 | 4.47 |
| Bass           | 3.76     | 3.37 | 4.58 | 6.90     | 6.77 | 3.36 | 7.96     | 7.99  | 5.74 |
| Others         | 3.19     | 2.36 | 2.85 | 5.00     | 4.40 | 2.65 | 9.35     | 9.04  | 3.60 |
| Vocal          | 6.71     | 5.95 | 3.97 | 7.61     | 6.98 | 3.59 | 15.55    | 14.70 | 4.56 |
| Entire dataset | 4.54     | 4.42 | 4.13 | 6.46     | 6.32 | 3.49 | 12.20    | 11.59 | 5.59 |



#### 4.4. Comparison of the SDR Metric with Existing Solutions

In the final model, there is a noticeable quantitative leap compared to the baseline model; for some paths, the median SDR increase is almost twofold. However, a comparison of the final result with solutions developed by other authors will be more meaningful. Therefore, Table 11 compares the model developed in this study with existing solutions trained using the MUSDB18 dataset. The median SDR for each path was used as the metric because it is most widely used in the literature. The developed model outperforms U-Net-based architectures operating on spectrograms, including C-U-Net and M-U-Net. In this category, only the spleeter model demonstrates an advantage. Compared with architectures that use recurrent layers, such as Open Umix and MMdenseLSTM, the developed solution is inferior for instrument separation but performs better for vocal separation. In the category of time-domain models, the developed solution is superior to wave-U-Net in all respects, but inferior to the Demucs model.

The U-Net presented in the article is a typical structure created from convolutional elements and can be used for many tasks. The Demucs model, on the other hand, incorporates a bidirectional LSTM layer between the encoding and decoding parts. Therefore, the use of a typical U-Net is not only more versatile compared to the Demucs models, but the training process is also less demanding. The Demucs models were trained with eight V100 GPUs with 16 GB of RAM, while the developed U-Net was trained in a Google Colab environment, using a single NVIDIA L4 GPU with 24 GB of VRAM.

The developed model produces the best results in vocal separation, outperforming most of the compared solutions. On instrument separation tasks, the developed model is outperformed by solutions using recurrent networks.

**Table 11.** Comparison of the SDR metric with existing solutions.

| Model            | Drums | Bass | Others | Vocal |
|------------------|-------|------|--------|-------|
| Baseline Model   | 3.63  | 1.79 | 1.57   | 4.85  |
| Final Model      | 5.66  | 3.76 | 3.19   | 6.71  |
| C-U-Net [15]     | 4.38  | 2.60 | 1.71   | 4.65  |
| M0U-Net [9]      | 4.89  | 2.99 | 2.58   | 5.44  |
| Spleeter [17]    | 5.93  | 5.10 | 4.24   | 6.55  |
| Wave-U-Net [16]  | 4.15  | 2.91 | 2.03   | 3.00  |
| Open Umix [19]   | 5.73  | 5.23 | 4.02   | 6.32  |
| MMdenseLSTM [20] | 6.62  | 5.19 | 4.93   | 4.94  |
| Demucs [24]      | 6.86  | 7.01 | 4.42   | 6.84  |

## 5. Conclusions and Future Work

The goal of this research was to develop a U-Net-based model for music source separation and to systematically analyze how architectural and training modifications influence its performance. Starting from a baseline multi-channel U-Net, successive adjustments were introduced and evaluated, including normalization, optimizer selection, regularization, dropout configuration, loss weighting strategies, data augmentation, spectrogram-domain sampling, and Wiener filtering.

The experiments show that substantial improvements can be achieved without modifying the core convolutional architecture. The median SDR for the entire dataset increased from 2.99 dB in the baseline model to 4.54 dB in the final configuration, with particularly strong gains for vocals and drums. The most significant improvement resulted from computing the loss on spectrograms transformed back to the original linear frequency

scale, despite training in logarithmic space. Additional consistent gains were obtained through batch normalization, energy-based loss weighting, increased dropout in the decoder, and time–frequency augmentation. A single-iteration Wiener filter further improved interference suppression, at the cost of additional artifacts.

Comparison with existing MUSDB18-based systems indicates that the optimized U-Net outperforms earlier spectrogram-based U-Net variants and achieves competitive vocal separation results, while remaining inferior to hybrid and recurrent architectures in instrument separation, especially for bass and complex accompaniment. The performance ceiling is primarily constrained by the purely convolutional design and reliance on magnitude-only spectrograms.

Future development may therefore involve extending the architecture with recurrent or transformer-based components to better model temporal dependencies, incorporating phase-aware or hybrid time–frequency representations, and expanding the training dataset to improve robustness across diverse musical styles. Overall, the study demonstrates that careful optimization of training procedures, preprocessing strategies and proper data augmentation can significantly extend the practical performance limits of a standard U-Net for music source separation, but architectural innovations are required to match the performance of state-of-the-art hybrid systems.

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## References

1. Cardoso, J.F. Blind signal separation: Statistical principles. *Proc. IEEE* **1998**, *86*, 2009–2025. [CrossRef]
2. Hyvärinen, A.; Hurri, J.; Hoyer, P.O. Independent component analysis. In *Natural Image Statistics: A Probabilistic Approach to Early Computational Vision*; Springer: Berlin/Heidelberg, Germany, 2001; pp. 151–175.
3. Smaragdis, P.; Raj, B.; Sashanka, M. Supervised and semi-supervised separation of sounds from single-channel mixtures. In *Proceedings of the International Conference on Independent Component Analysis and Signal Separation*; Springer: Berlin/Heidelberg, Germany, 2007; pp. 414–421.
4. Cañadas-Quesada, F.J.; FitzGerald, D.; Vera-Candeas, P.; Ruiz-Reyes, N. Harmonic-percussive sound separation using rhythmic information from non-negative matrix factorization in single-channel music recordings. In *Proceedings of the International Conference on Digital Audio Effects (DAFx)*, Edinburgh, UK, 5–9 September 2017; pp. 276–282.

5. Rafii, Z.; Pardo, B. Repeating pattern extraction technique (REPET): A simple method for music/voice separation. *IEEE Trans. Audio Speech Lang. Process.* **2012**, *21*, 73–84. [\[CrossRef\]](#)
6. Araki, S.; Ito, N.; Haeb-Umbach, R.; Wichern, G.; Wang, Z.Q.; Mitsufuji, Y. 30+ years of source separation research: Achievements and future challenges. In Proceedings of the ICASSP 2025–2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Hyderabad, India, 6–11 April 2025; pp. 1–5.
7. Gul, S.; Khan, M.S. A survey of audio enhancement algorithms for music, speech, bioacoustics, biomedical, industrial, and environmental sounds by image U-Net. *IEEE Access* **2023**, *11*, 144456–144483. [\[CrossRef\]](#)
8. Rafii, Z.; Liutkus, A.; Stöter, F.R.; Mimilakis, S.I.; Bittner, R. The MUSDB18 Corpus for Music Separation. 2017. Available online: <https://zenodo.org/records/1117372> (accessed on 12 December 2025).
9. Kadandale, V.S.; Montesinos, J.F.; Haro, G.; Gómez, E. Multi-channel u-net for music source separation. In Proceedings of the 2020 IEEE 22nd International Workshop on Multimedia Signal Processing (MMSP), Tampere, Finland, 21–24 September 2020; pp. 1–6.
10. Ronneberger, O.; Fischer, P.; Brox, T. U-Net: Convolutional Networks for Biomedical Image Segmentation. In *Proceedings of the Medical Image Computing and Computer-Assisted Intervention—MICCAI 2015*; Navab, N., Hornegger, J., Wells, W.M., Frangi, A.F., Eds.; Springer International Publishing: Cham, Switzerland, 2015; pp. 234–241.
11. Vincent, E.; Gribonval, R.; Fevotte, C. Performance measurement in blind audio source separation. *IEEE Trans. Audio Speech Lang. Process.* **2006**, *14*, 1462–1469. [\[CrossRef\]](#)
12. Peeters, G. Spectral and temporal periodicity representations of rhythm for the automatic classification of music audio signal. *IEEE Trans. Audio Speech Lang. Process.* **2010**, *19*, 1242–1252. [\[CrossRef\]](#)
13. Su, L.; Yang, Y.H. Combining spectral and temporal representations for multipitch estimation of polyphonic music. *IEEE/ACM Trans. Audio Speech Lang. Process.* **2015**, *23*, 1600–1612. [\[CrossRef\]](#)
14. Siddique, N.; Sidike, P.; Elkin, C.; Devabhaktuni, V. U-Net and its variants for medical image segmentation: Theory and applications. *arXiv* **2020**, arXiv:2011.01118. [\[CrossRef\]](#)
15. Meseguer-Brocal, G.; Peeters, G. Conditioned-U-Net: Introducing a control mechanism in the U-Net for multiple source separations. *arXiv* **2019**, arXiv:1907.01277.
16. Stoller, D.; Ewert, S.; Dixon, S. Wave-u-net: A multi-scale neural network for end-to-end audio source separation. *arXiv* **2018**, arXiv:1806.03185.
17. Hennequin, R.; Khelif, A.; Voituret, F.; Moussallam, M. Spleeter: A fast and efficient music source separation tool with pre-trained models. *J. Open Source Softw.* **2020**, *5*, 2154. [\[CrossRef\]](#)
18. Sawant, O.; Bhowmick, A.; Bhagwat, G. Separation of speech & music using temporal-spectral features and neural classifiers. *Evol. Intell.* **2024**, *17*, 1389–1403.
19. Stöter, F.R.; Uhlich, S.; Liutkus, A.; Mitsufuji, Y. Open-unmix-a reference implementation for music source separation. *J. Open Source Softw.* **2019**, *4*, 1667. [\[CrossRef\]](#)
20. Takahashi, N.; Goswami, N.; Mitsufuji, Y. Mmdenselstm: An efficient combination of convolutional and recurrent neural networks for audio source separation. In Proceedings of the 2018 16th International Workshop on Acoustic Signal Enhancement (IWAENC), Tokyo, Japan, 17–20 September 2018; pp. 106–110.
21. Wang, C.H.; Wang, C.C.; Wang, J.Y.; Jang, J.S.R.; Chu, Y.H. Improving Real-Time Music Accompaniment Separation with MMDenseNet. In Proceedings of the 2024 27th Conference of the Oriental COCODA International Committee for the Co-ordination and Standardisation of Speech Databases and Assessment Techniques (O-COCODA), Hsinchu, Taiwan, 17–19 October 2024; pp. 1–6.
22. Liutkus, A.; Stöter, F.R.; Rafii, Z.; Kitamura, D.; Rivet, B.; Ito, N.; Ono, N.; Fontecave, J. The 2016 Signal Separation Evaluation Campaign. In *Proceedings of the Latent Variable Analysis and Signal Separation—12th International Conference, LVA/ICA 2015, Liberec, Czech Republic, 25–28 August 2015, Proceedings*; Tichavský, P., Babaie-Zadeh, M., Michel, O.J., Thirion-Moreau, N., Eds.; Springer International Publishing: Cham, Switzerland, 2017; pp. 323–332.
23. Takahashi, N.; Mitsufuji, Y. D3net: Densely connected multidilated densenet for music source separation. *arXiv* **2020**, arXiv:2010.01733.
24. Défossez, A.; Usunier, N.; Bottou, L.; Bach, F. Music source separation in the waveform domain. *arXiv* **2019**, arXiv:1911.13254.
25. Rafii, Z.; Liutkus, A.; Stöter, F.R.; Mimilakis, S.I.; Bittner, R. MUSDB18-HQ—An Uncompressed Version of MUSDB18. 2019. Available online: <https://zenodo.org/records/3338373> (accessed on 12 December 2025). [\[CrossRef\]](#)
26. Luo, Y.; Yu, J. Music source separation with band-split RNN. *IEEE/ACM Trans. Audio Speech Lang. Process.* **2023**, *31*, 1893–1901. [\[CrossRef\]](#)
27. Lu, W.T.; Wang, J.C.; Kong, Q.; Hung, Y.N. Music source separation with band-split rope transformer. In Proceedings of the ICASSP 2024–2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Seoul, Republic of Korea, 14–19 April 2024; pp. 481–485.
28. Wang, J.C.; Lu, W.T.; Won, M. Mel-band RoFormer for music source separation. *arXiv* **2023**, arXiv:2310.01809.

29. Takahashi, N.; Mitsufuji, Y. Multi-scale multi-band densenets for audio source separation. In Proceedings of the 2017 IEEE Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA), New Paltz, NY, USA, 15–18 October 2017; pp. 21–25.
30. Chen, J.; Vekkot, S.; Shukla, P. Music source separation based on a lightweight deep learning framework (DTTNET: Dual-path TFC-TDF UNet). In Proceedings of the ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Seoul, Republic of Korea, 14–19 April 2024; pp. 656–660.
31. Daly, M. Remixing music for hearing aids using ensemble of fine-tuned source separators. In Proceedings of the 2024 IEEE International Conference on Acoustics, Speech, and Signal Processing Workshops (ICASSPW), Seoul, Republic of Korea, 14–19 April 2024; pp. 109–110.
32. Jeon, C.B.; Wichern, G.; Germain, F.G.; Le Roux, J. Why does music source separation benefit from cacophony? In Proceedings of the 2024 IEEE International Conference on Acoustics, Speech, and Signal Processing Workshops (ICASSPW), Seoul, Republic of Korea, 14–19 April 2024; pp. 873–877.
33. Khera, H.; Pauwels, J.; Archer-Boyd, A.W.; Sandler, M.B. Beyond Architecture: The Critical Impact of Inference Overlap on Music Source Separation Benchmarks. In Proceedings of the 2025 IEEE Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA), Tahoe City, CA, USA, 12–15 October 2025.
34. Stöter, F.R.; Liutkus, A.; Ito, N. The 2018 signal separation evaluation campaign. In *Proceedings of the International Conference on Latent Variable Analysis and Signal Separation*; Springer: Berlin/Heidelberg, Germany, 2018; pp. 293–305.
35. Bittner, R.M.; Wilkins, J.; Yip, H.; Bello, J.P. MedleyDB 2.0: New data and a system for sustainable data collection. *ISMIR Late Break. Demo Pap.* **2016**, 36.
36. Harris, F.J. On the use of windows for harmonic analysis with the discrete Fourier transform. *Proc. IEEE* **2005**, *66*, 51–83. [[CrossRef](#)]
37. Jansson, A.; Humphrey, E.; Montecchio, N.; Bittner, R.; Kumar, A.; Weyde, T. Singing voice separation with deep u-net convolutional networks. In Proceedings of the 18th International Society for Music Information Retrieval Conference, Suzhou, China, 23–27 October 2017.
38. Luo, Y.; Mesgarani, N. Tasnet: Time-domain audio separation network for real-time, single-channel speech separation. In Proceedings of the 2018 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Calgary, AB, Canada, 15–20 April 2018; pp. 696–700.
39. Cohen-Hadria, A.; Roebel, A.; Peeters, G. Improving singing voice separation using deep u-net and wave-u-net with data augmentation. In Proceedings of the 2019 27th European Signal Processing Conference (EUSIPCO), A Coruña, Spain, 2–6 September 2019; pp. 1–5.
40. Wang, H.; Zou, Y.; Wang, W.S. A hidden space data augmentation method for acoustic scene classification. *arXiv* **2021**, arXiv:2103.16858. [[CrossRef](#)]
41. Fitzgerald, D.; Jaiswal, R. On the use of masking filters in sound source separation. In Proceedings of the 15th International Conference on Digital Audio Effects (DAFx-12), York, UK, 17–21 September 2012.

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